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Introduction:

User Interface (UI) design focuses on creating interactive systems that are easy for users to understand and operate. One of the important steps in UI design is **Task Flow Analysis**, which helps designers understand how users interact with a system to complete a specific task. By analyzing the sequence of actions performed by users, designers can create efficient interfaces that improve usability and user experience.

In this experiment, a **Coffee Shop Workflow** is used as an example to demonstrate task flow analysis and wireframe representation. The workflow shows how a customer interacts with the coffee shop system from entering the shop to completing the order and exiting the shop.

The wireframe and workflow diagrams are designed using **Figma**,**Lucid chart** which helps designers visually structure interfaces before actual development.

Task Flow Analysis:

Task Flow Analysis describes the **step-by-step sequence of actions** that a user performs to complete a task within a system. It helps identify the logical order of operations and decision points in a process

For the coffee shop system, the task flow includes the following steps:

1. Customer enters the coffee shop.
2. Customer views the menu.
3. Customer selects coffee or other beverages.
4. Customer chooses **Dine-In or Takeaway** option.
5. Customer proceeds to payment.
6. System processes the payment.

Decision points in the workflow include:

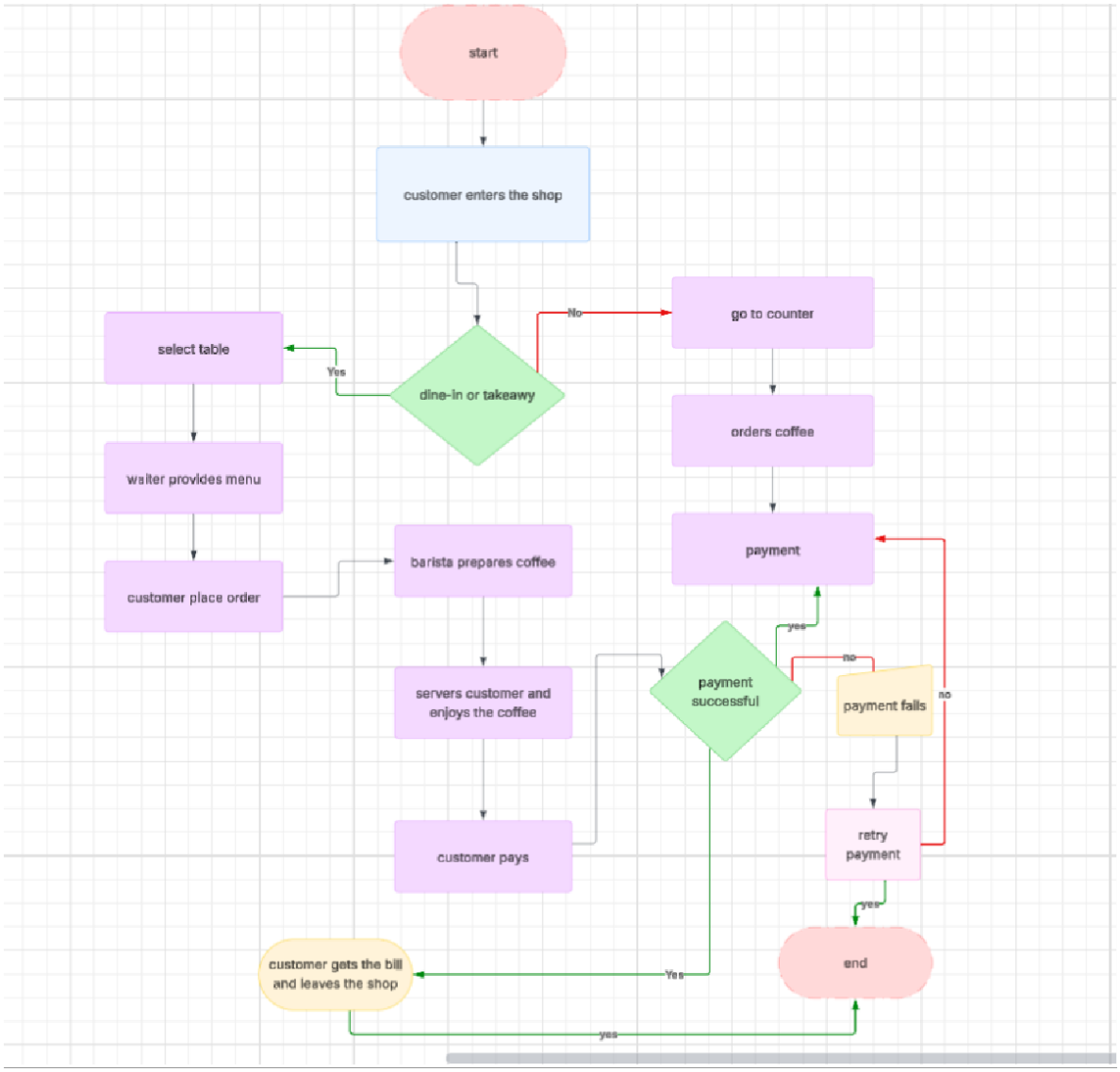
- If **payment is successful**, the order is confirmed.
- If **payment fails**, the system notifies the customer and requests another payment attempt.

After payment confirmation:

- If the customer chooses **Dine-In**, they receive the order and enjoy their coffee in the shop.
- If the customer chooses **Takeaway**, the order is packed and the customer exits the shop.

Task Flow Analysis helps in understanding how users move through the system and ensures that the interface supports these actions effectively.

DIAGRAMATIC WORK FLOW:



Wireframe Representation of Coffee Shop Workflow

A wireframe is a **basic visual layout of an interface** that shows the placement of elements such as buttons, menus, text fields, and navigation components.
Wireframes focus on structure and functionality rather than colors or detailed design.

For the coffee shop workflow, the wireframe represents the following screens:

- **Welcome Screen** – Displays the coffee shop interface and entry point.
- **Menu Screen** – Shows available coffee items and beverages.
- **Order Selection Screen** – Allows customers to select items and quantity.
- **Order Type Screen** – Option to choose between Dine-In or Takeaway.
- **Payment Screen** – Displays payment options.
- **Payment Status Screen** – Shows whether the payment is successful or failed.
- **Order Confirmation Screen** – Confirms order and prepares coffee.
- **Completion Screen** – Customer receives order and exits the shop.

Wireframes help designers visualize the structure of the system and ensure that the navigation between screens is clear and logical.

Framework/Tool Used

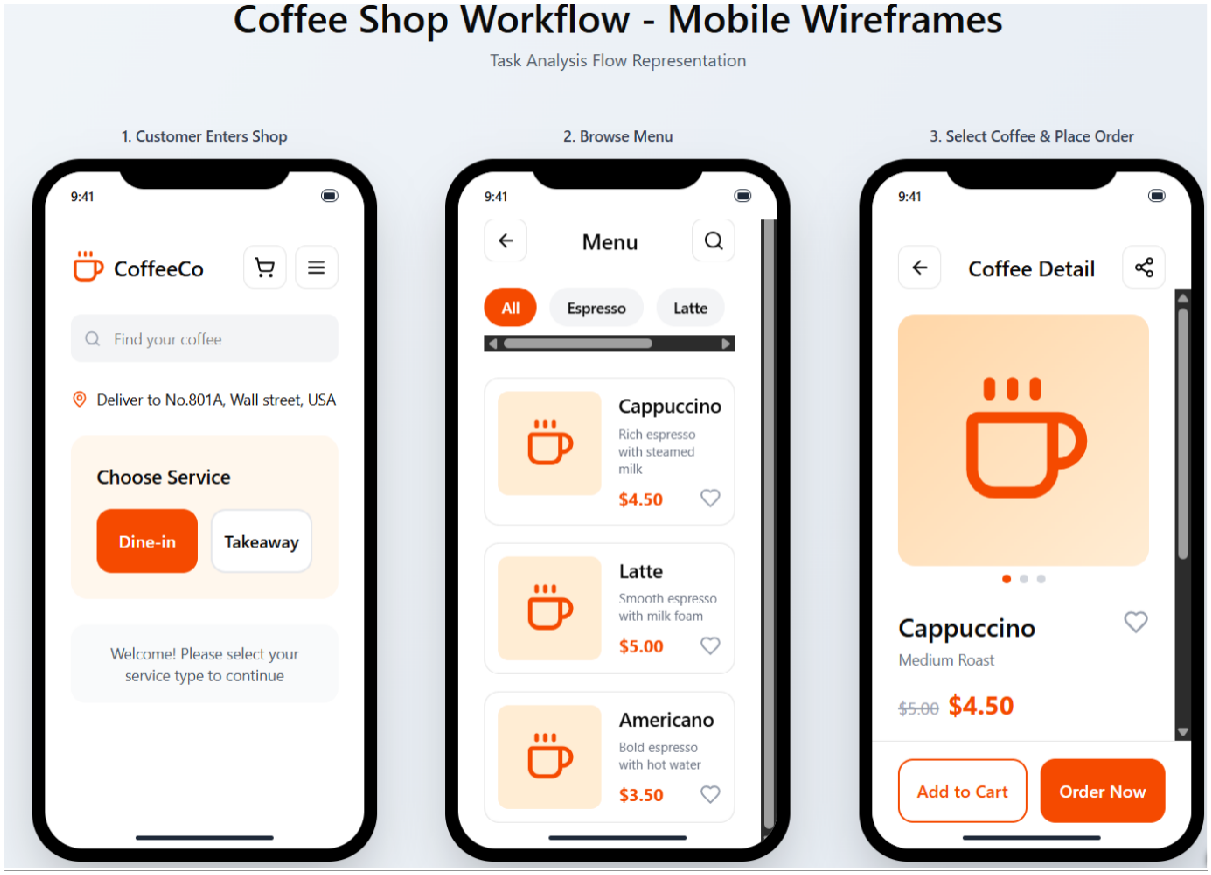
The wireframe and workflow diagrams were created using **Figma**.

Figma is a cloud-based design and prototyping tool widely used for UI and UX design. It allows designers to create layouts, wireframes, and interactive prototypes collaboratively.

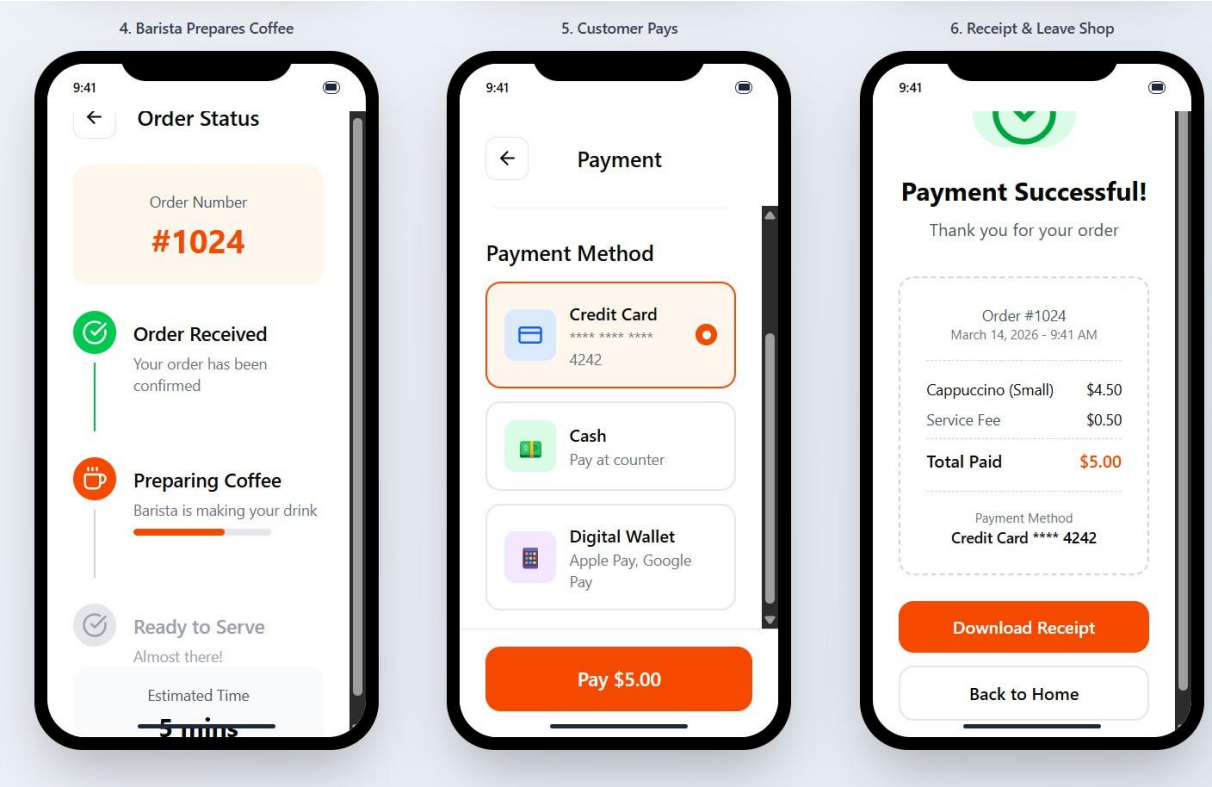
Key advantages of Figma include:

- Real-time collaboration
- Easy prototyping
- Cloud-based access
- Drag-and-drop design components

Framework representations:



As the following frameworks were taken from figma it very efficient way to understand how frameworks works buy this means of way user can understand what all are the required steps to take measurement accordingly and moves to the next step by this way user can estimate and understand what to do (guides) the user to be more approachable and main aim of this framework was to give an better experience and give out an structured formatting this way user interface archives the goal of formatting and user's intuitive experience.



Thereby we can say that wireframe has UI/UX interface which allows the user to be intuitive and more realistic way of understanding and giving an valuable user experience. They help maintain consistency and organization during the development process. Using a framework reduces development time and improves productivity. It also ensures that applications follow a standard architecture and best practices. Frameworks support easier debugging, testing, and maintenance of projects. They allow developers to focus more on functionality rather than basic setup. Overall, frameworks make the development process faster, more reliable, and easier to manage.

Features of the Coffee Shop UI Design:

The coffee shop workflow interface includes the following features:

- **Simple Navigation** – Easy movement between menu, order, and payment screens.
- **Clear Menu Display** – Organized view of available coffee and beverages.
- **Order Customization** – Users can select their preferred items and quantities.
- **Dine-In and Takeaway Options** – Flexible ordering based on customer preference.
- **Secure Payment Processing** – Handles payment verification and confirmation.
- **Payment Failure Notification** – Alerts users if payment is unsuccessful.
- **Order Confirmation System** – Confirms and prepares the selected order.
- **User-Friendly Interface** – Designed for quick and easy interaction.

Conclusion

Task Flow Analysis and wireframe representation are essential steps in designing effective user interfaces. They help designers understand how users interact with the system and ensure that the interface supports smooth navigation and task completion.

Using **Figma**, the coffee shop workflow was structured and visualized through wireframes, allowing the design to be organized before implementation. This approach improves usability, reduces design errors, and enhances the overall user experience.